



Computer Vision for Autonomous Vehicles

Correspondences and RANSAC

Instructor: Golnaz Habibi, PhD

University of Oklahoma

Fall 2025



Announcement

- Lab 5 posted. Due this Friday
- Final project proposal due is November 3rd
- Deadline for undergraduate section to notify of their interest: Oct 23rd! (up to additional %5 of their final grade)
- We will have two guest lecturer
 - First lecture: Friday Oct 31st, REPF 200
 - by Dr. Jonathan How from MIT
 - No class on Oct 30th. But attendance is mandatory in this lecture, please sign up and take a picture from the talk
 - Takeaway quiz
- Today:
 - Pose estimation under uncertainty
 - Lab 5 Review, may have a quiz



**JONATHAN
HOW, Ph.D.**

*Ford Professor of
Engineering*

*Massachusetts Institute
of Technology*

RESILIENT AUTONOMY

Perception and Planning for Dynamic, Unknown Environments

Presented by Jonathan How, PhD

ABSTRACT

Unmanned aerial systems (UAS) have the potential to transform domains such as search and rescue, environmental monitoring, and package delivery. However, deploying UAS in real-world settings poses fundamental challenges: operating in GPS-denied environments, reducing uncertainty

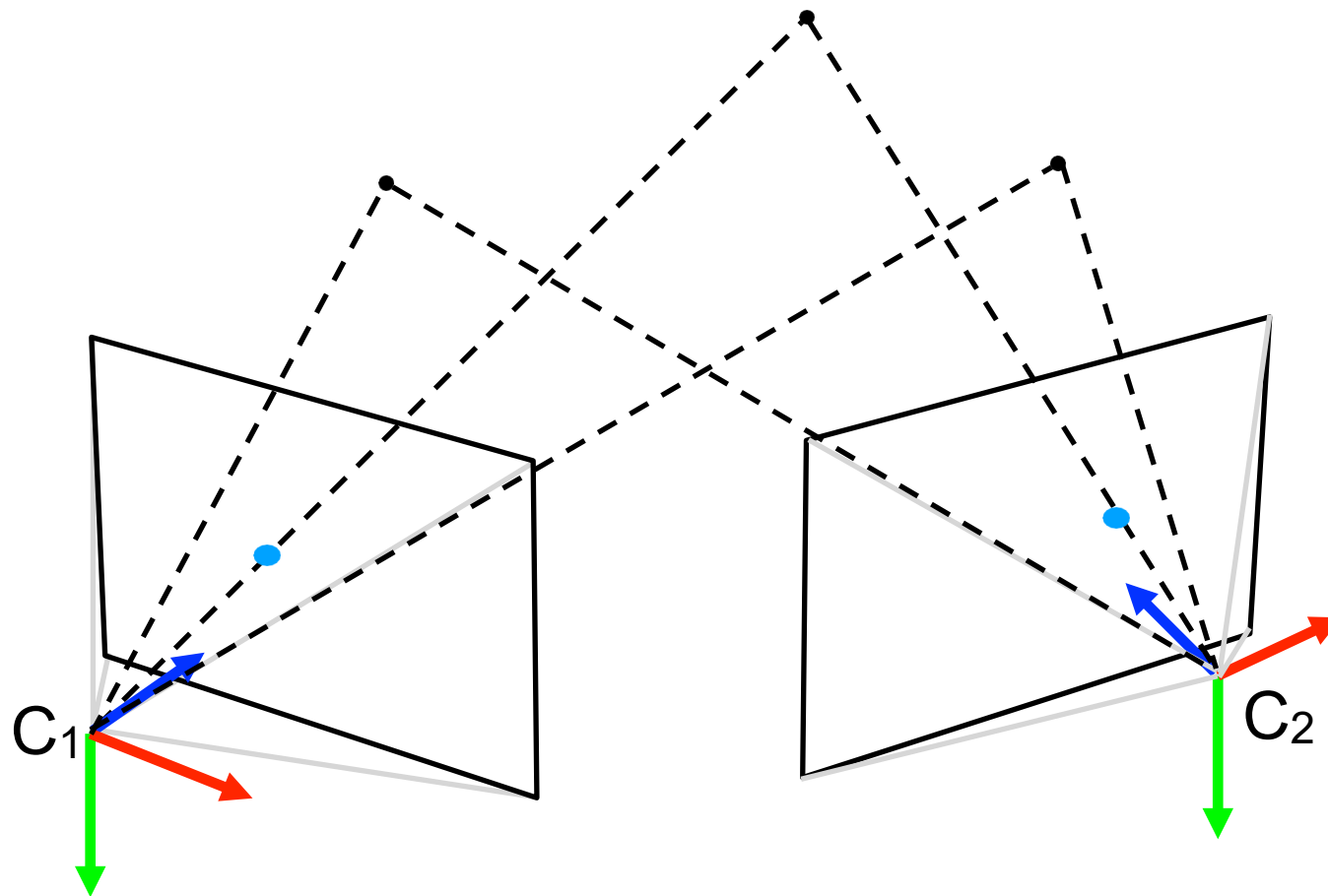
Friday, October 31

2:30-4 pm

Rawl Engineering Practice Facility,
Room 200

relocalization and loop closures under extreme viewpoint changes. On the planning side, we introduce two complementary frameworks that leverage these perception capabilities. **DYNUS** is an uncertainty-aware trajectory planner designed for safe flight in dynamic, unknown environments.

2-view Geometry from 3D-3D Correspondences

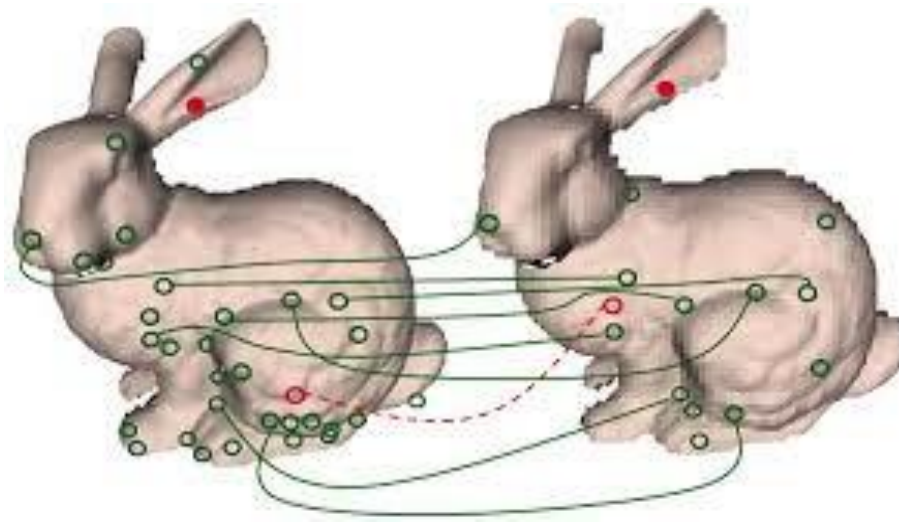


How to estimate the relative pose between the cameras from 3D-3D correspondences $(p_{1,k}, p_{2,k})$ with $k = 1, \dots, N$?

Few More Comments:

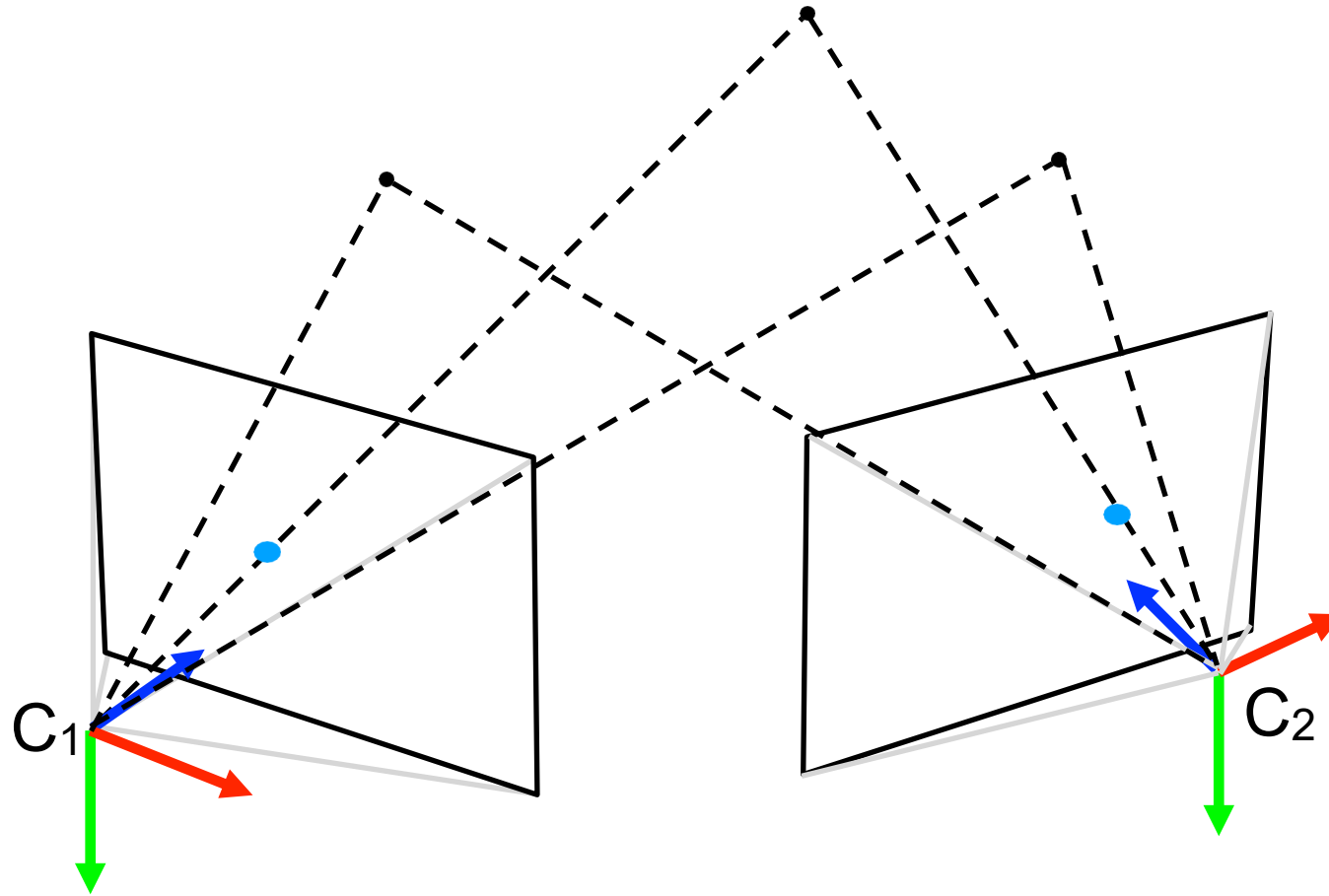
- **3 points** are sufficient to compute the relative pose from 3D-3D correspondences
 - We can use the solver seen today as a 3-point minimal solver within a **RANSAC** method
- **Also useful for 3D objects localization:**

Other names: vector registration, point cloud alignment, ..



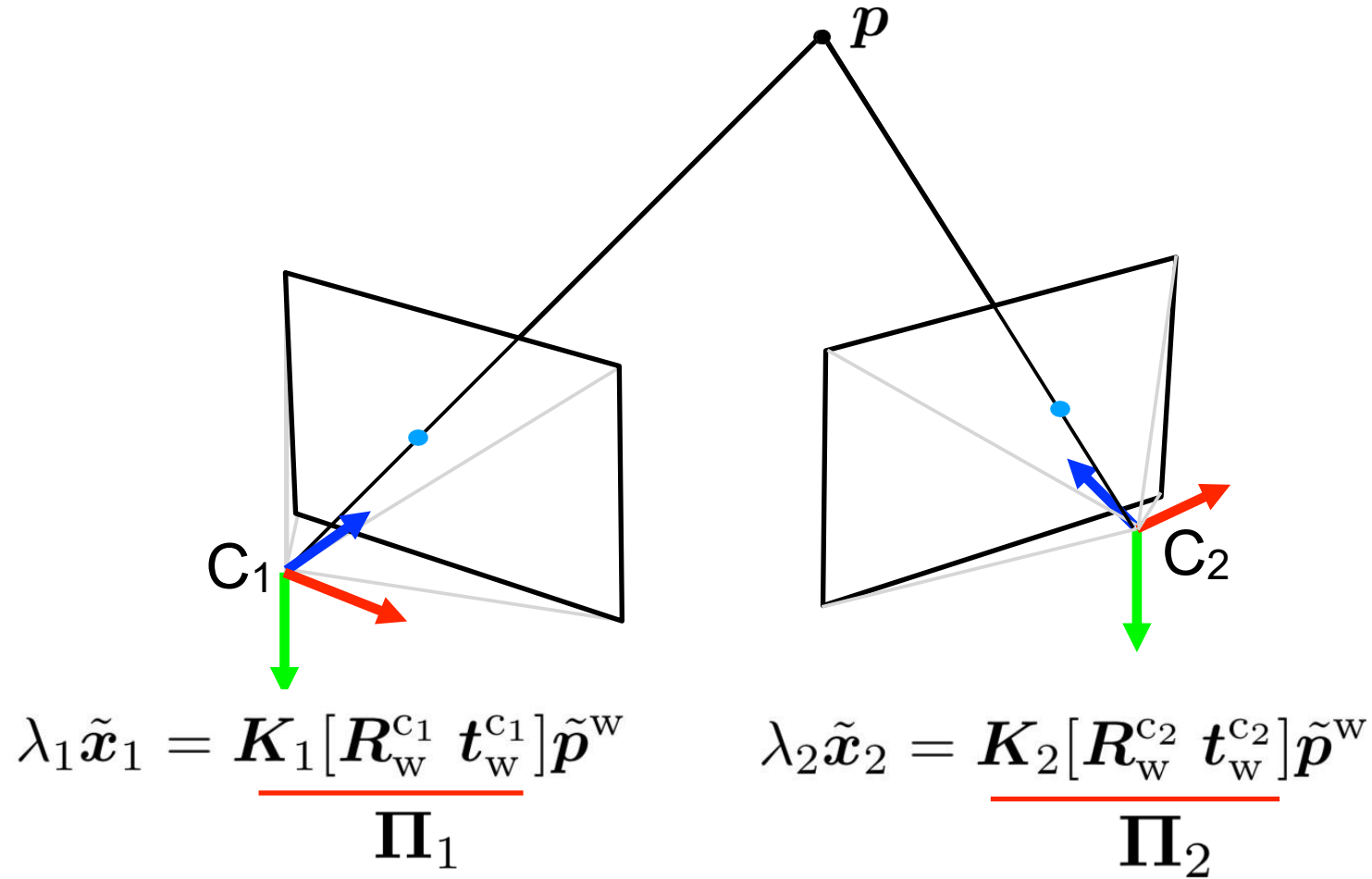
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Recap: 2-view Geometry from 3D-3D Correspondences



**How to estimate the relative pose between the cameras
from 3D-3D correspondences $(p_{1,k}, p_{2,k})$ with $k = 1, \dots, N$?**

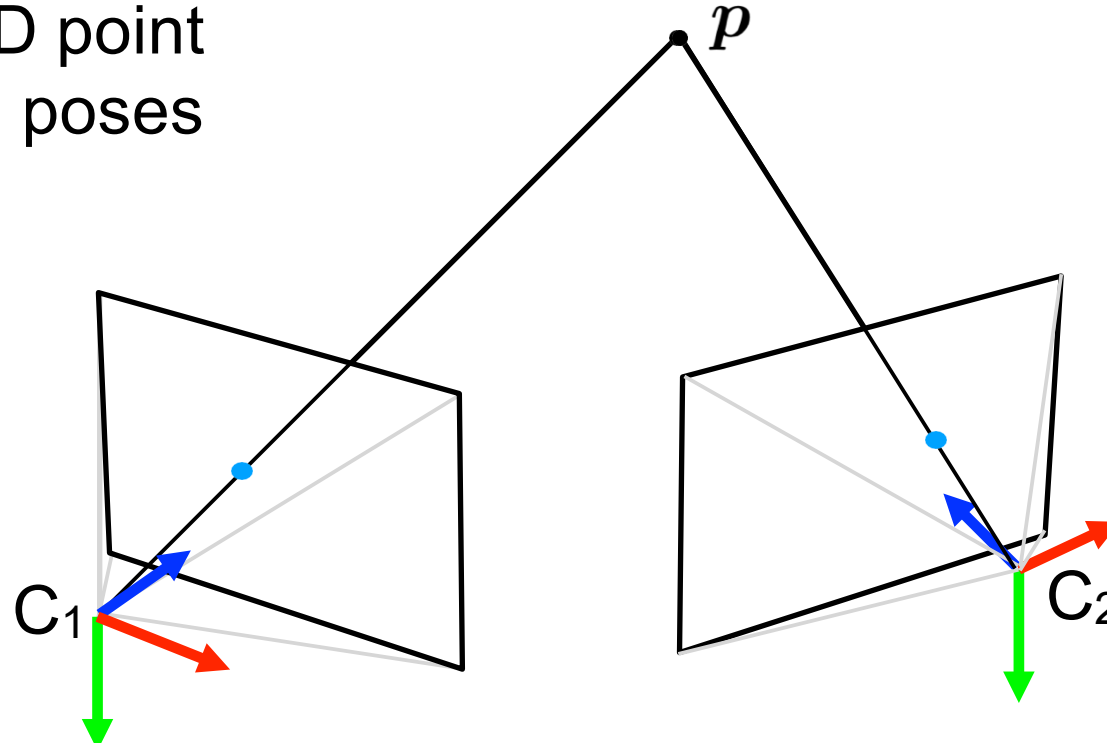
Example **1a**: Triangulation (Structure Reconstruction)
 Compute 3D point from known poses



Linear triangulation: $\min_{\|\tilde{p}^w\|=1} \|A\tilde{p}^w\|^2$

Example **1b**: Triangulation (Structure Reconstruction)

Compute 3D point
from known poses

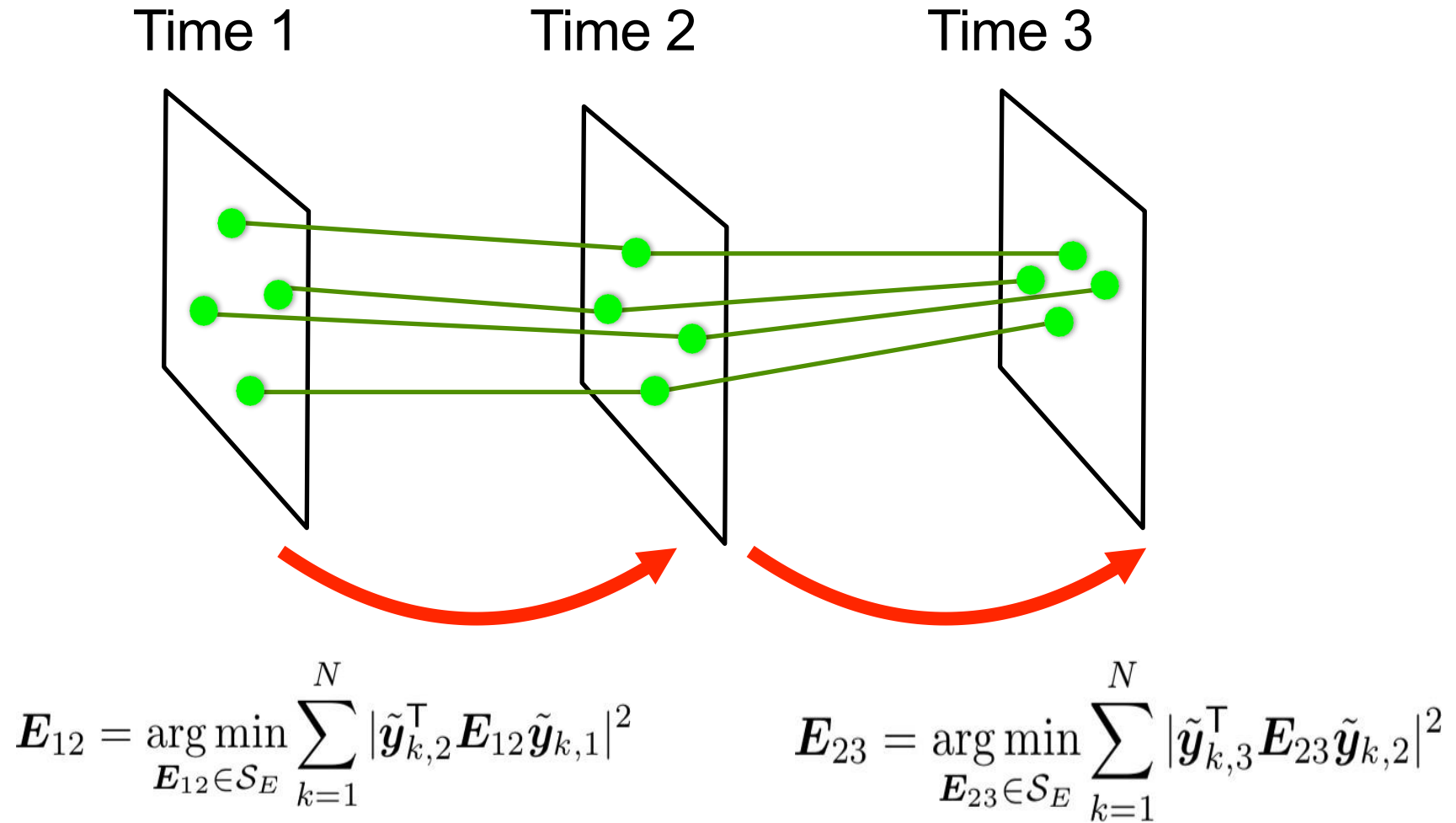


$$\lambda_1 \tilde{x}_1 = \underbrace{K_1 [R_w^{c_1} \ t_w^{c_1}]}_{\Pi_1} \tilde{p}^w$$

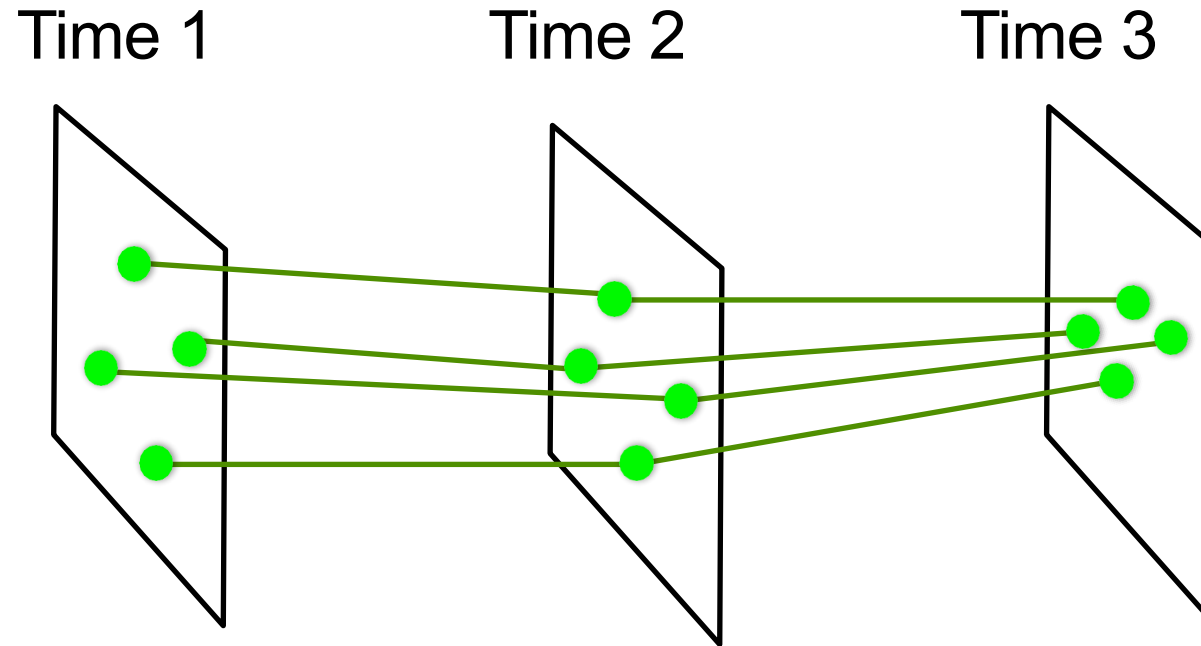
$$\lambda_2 \tilde{x}_2 = \underbrace{K_2 [R_w^{c_2} \ t_w^{c_2}]}_{\Pi_2} \tilde{p}^w$$

$$\min_{p^w} \|x_1 - \pi(R_{c_1}^w, t_{c_1}^w, p^w)\|^2 + \|x_2 - \pi(R_{c_2}^w, t_{c_2}^w, p^w)\|^2$$

Example 2a: Motion Estimation



Example 2b: Motion Estimation



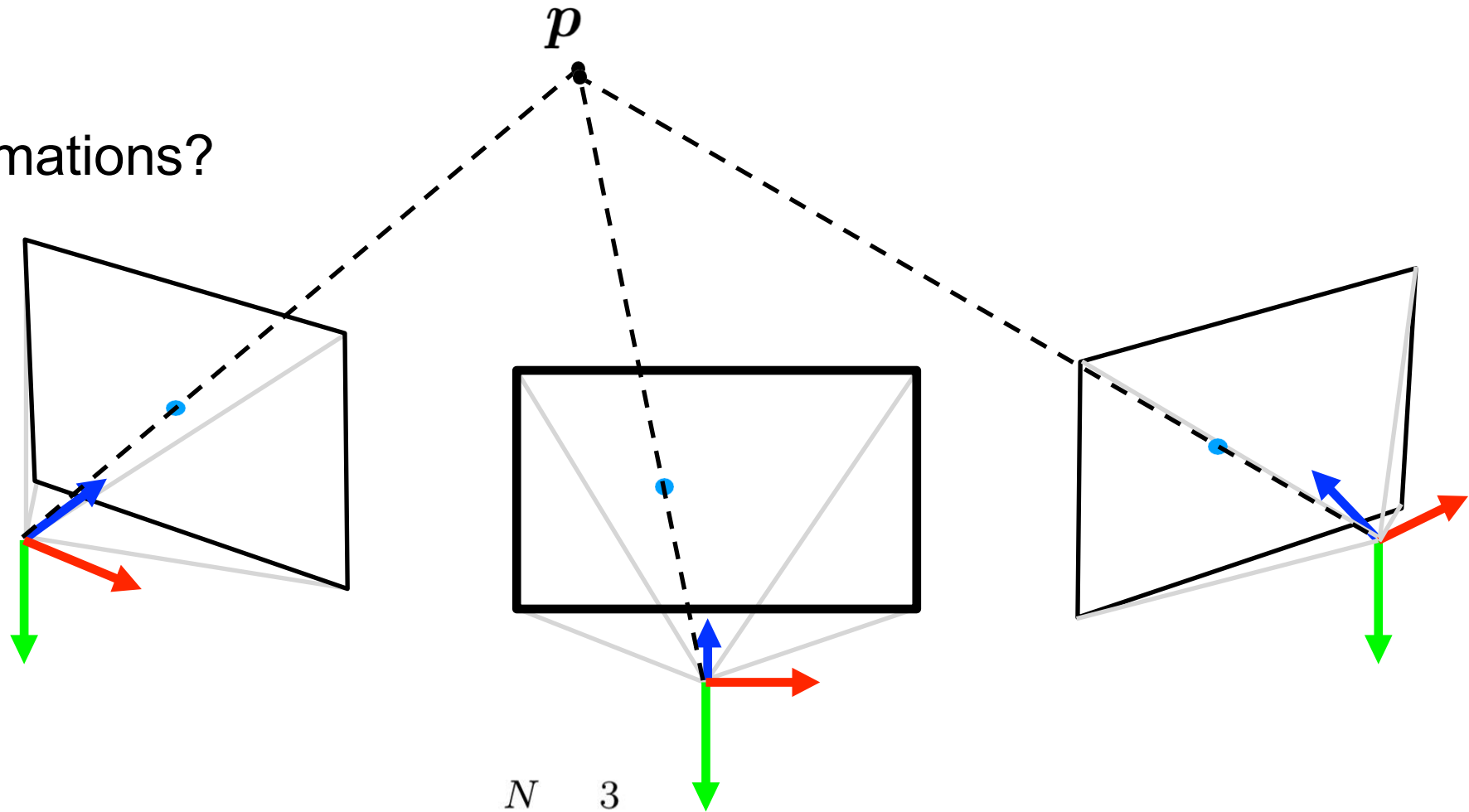
$$\min_{\substack{(\mathbf{R}_{c_i}^w, \mathbf{t}_{c_i}^w), i=1,2,3 \\ \mathbf{p}_k^w, k=1,\dots,N}} \sum_{k=1}^N \sum_{i=1}^3 \|\mathbf{x}_{k,i} - \pi(\mathbf{R}_{c_i}^w, \mathbf{t}_{c_i}^w, \mathbf{p}_k^w)\|^2$$

Generalizes to K cameras: **Bundle adjustment**

Example 2b: Motion and Structure Estimation

Scale?

Rigid transformations?



$$\min_{\substack{(\mathbf{R}_{c_i}^w, \mathbf{t}_{c_i}^w), i=1,2,3 \\ \mathbf{p}_k^w, k=1, \dots, N}} \sum_{k=1}^N \sum_{i=1}^3 \|\mathbf{x}_{k,i} - \pi(\mathbf{R}_{c_i}^w, \mathbf{t}_{c_i}^w, \mathbf{p}_k^w)\|^2$$

Generalizes to K cameras: **Bundle adjustment**

Estimation Theory

Concerned with the estimation of unknown variables given (noisy) measurements and prior information

Estimator: a function of the measurements that approximates the unknown variables

Measurements that depend on some unknown variable $\mathbf{x}$:

$$z_1, \dots, z_N$$



Estimator for $\mathbf{x}$:

$$x^* = \mathcal{F}(z_1, \dots, z_N)$$

$$x^* \approx x$$



Maximum Likelihood

Estimation (MLE)

Assume we are given N measurements $z_1, \dots, z_N$ (e.g., pixel measurements) that are function of a variable we want to estimate $\mathbf{x}$ (e.g., camera poses, points). Assume that we are also given the conditional distributions:

$$\mathbb{P}(z_j|\mathbf{x})$$

Then the *maximum likelihood* estimator (MLE) is defined as:

$$\mathbf{x}_{\text{MLE}} = \arg \max_{\mathbf{x}} \mathbb{P}(\underline{z_1, \dots, z_N}|\mathbf{x})$$

Measurement
likelihood

where $\mathbb{P}(z_1, \dots, z_N|\mathbf{x})$ is also called the *likelihood* of the measurements given $\mathbf{x}$. Equivalently:

$$\mathbf{x}_{\text{MLE}} = \arg \min_{\mathbf{x}} \underline{-\log \mathbb{P}(z_1, \dots, z_N|\mathbf{x})}$$

Negative
log-likelihood

Maximum Likelihood Estimation (MLE)

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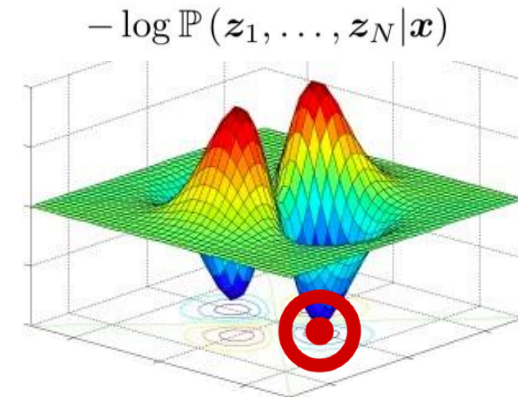
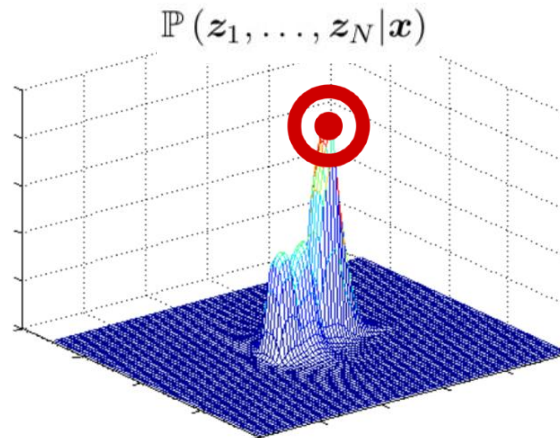
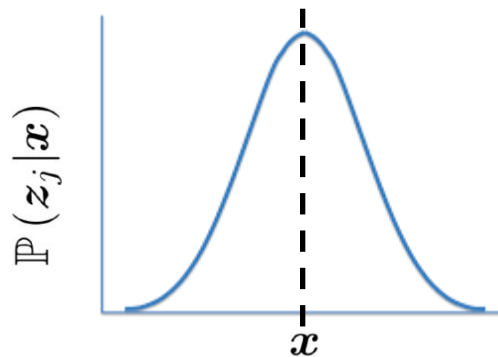
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Measurement
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$$\mathbf{x}_{\text{MLE}} = \arg \min_{\mathbf{x}} -\log \mathbb{P}(z_1, \dots, z_N | \mathbf{x})$$

Negative
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Maximum a Posteriori

Estimation (MAP)

Assume we are given N measurements $z_1, \dots, z_N$ (e.g., pixel measurements) that are function of a variable we want to estimate $\mathbf{x}$ (e.g., camera poses, points). *Maximum a Posteriori Estimation* (MAP) is a generalization of MLE. Then the MAP estimator is:

(15.30)

(15.31)

(15.32)

$$\arg \max_{\mathbf{x}} \underbrace{\mathbb{P}(z_1, \dots, z_N | \mathbf{x})}_{\text{Measurement likelihood}} \underbrace{\mathbb{P}(\mathbf{x})}_{\text{Priors}} \quad (15.33)$$

Maximum a Posteriori Estimation (MAP)

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$$\mathbf{x}_{\text{MAP}} = \arg \max_{\mathbf{x}} \mathbb{P}(\mathbf{x} | \mathbf{z}_1, \dots, \mathbf{z}_N) \quad (15.30)$$

Using Bayes rule:

$$\mathbf{x}_{\text{MAP}} = \arg \max_{\mathbf{x}} \mathbb{P}(\mathbf{x} | \mathbf{z}_1, \dots, \mathbf{z}_N) = \quad (15.31)$$

$$\arg \max_{\mathbf{x}} \frac{\mathbb{P}(\mathbf{z}_1, \dots, \mathbf{z}_N | \mathbf{x}) \mathbb{P}(\mathbf{x})}{\mathbb{P}(\mathbf{z}_1, \dots, \mathbf{z}_N)} = \quad (15.32)$$

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Measurement
likelihood Priors

Maximum a Posteriori Estimation (MAP)

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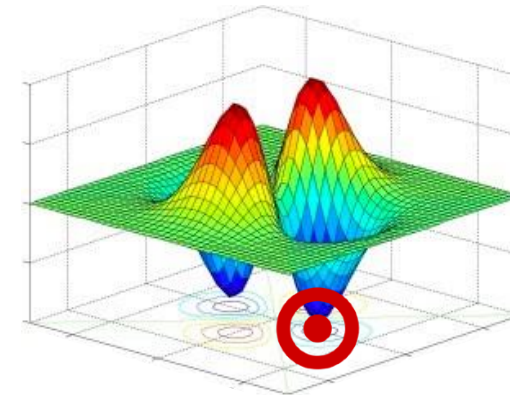
$$\mathbf{x}_{\text{MAP}} = \arg \max_{\mathbf{x}} \mathbb{P}(\mathbf{x} | z_1, \dots, z_N) = \quad (15.31)$$

$$\arg \max_{\mathbf{x}} \frac{\mathbb{P}(z_1, \dots, z_N | \mathbf{x}) \mathbb{P}(\mathbf{x})}{\mathbb{P}(z_1, \dots, z_N)} = \quad (15.32)$$

$$\arg \max_{\mathbf{x}} \underbrace{\mathbb{P}(z_1, \dots, z_N | \mathbf{x})}_{\text{Measurement likelihood}} \underbrace{\mathbb{P}(\mathbf{x})}_{\text{Priors}} \quad (15.33)$$

Assuming independence between measurements:

$$\mathbf{x}_{\text{MAP}} = \arg \min_{\mathbf{x}} - \sum_{j=1}^N \log \mathbb{P}(z_j | \mathbf{x}) - \log \mathbb{P}(\mathbf{x})$$



Optimization

Linear triangulation:

$$\min_{\|\tilde{\mathbf{p}}^w\|=1} \|\mathbf{A}\tilde{\mathbf{p}}^w\|^2$$



Nonlinear triangulation:

$$\begin{aligned} \min_{\mathbf{p}^w} & \|\mathbf{x}_1 - \pi(\mathbf{R}_{c_1}^w, \mathbf{t}_{c_1}^w, \mathbf{p}^w)\|^2 + \\ & + \|\mathbf{x}_2 - \pi(\mathbf{R}_{c_2}^w, \mathbf{t}_{c_2}^w, \mathbf{p}^w)\|^2 \end{aligned}$$

